AI, Machine Learning & Deep Learning

5 pages • foundations, learning paradigms, algorithms, neural networks and CNNs

1. AI, ML & DL Foundations

Artificial Intelligence (AI)

AI is the broad field of building systems that perform tasks associated with human intelligence, such as reasoning, perception, language understanding and decision-making.

Machine Learning (ML)

ML is a subset of AI in which systems learn patterns from data to make predictions or decisions instead of relying only on explicitly written rules.

Deep Learning (DL)

DL is a subset of ML that uses neural networks with multiple layers to learn complex representations, often from large datasets.

Relationship

AI ⊃ ML ⊃ DL. Every DL system is ML, but not every AI or ML system uses deep learning.

ML workflow

Define the problem → collect data → clean and explore → split data → preprocess → train a baseline → evaluate and tune → deploy → monitor.

Key terms

Feature = input variable; label/target = value to predict; training = learning parameters; inference = making predictions; generalization = performance on unseen data.

Key takeaway: Revision point: start with a clear problem and metric, then build a simple baseline before adding complexity.

2. Types of Machine Learning

Supervised learning

Learns from labelled examples. Regression predicts continuous values (e.g. house price); classification predicts classes (e.g. spam/not spam).

Unsupervised learning

Finds structure in unlabelled data. Clustering groups similar observations (K-Means, DBSCAN); dimensionality reduction compresses features (PCA).

Semi-supervised learning

Combines a small labelled dataset with a larger unlabelled dataset.

Self-supervised learning

Creates training targets from the data itself; widely used to pretrain language and vision models.

Reinforcement learning

An agent interacts with an environment, takes actions and learns from rewards or penalties.

Overfitting vs underfitting

Underfitting: model is too simple and performs poorly on training and test data. Overfitting: model memorizes training patterns/noise and performs poorly on unseen data.

Preventing leakage

Fit preprocessing only on training data; keep test data untouched until final evaluation. Leakage occurs when information unavailable at prediction time enters training.

Key takeaway: Metrics: regression—MAE, MSE, RMSE, R^2; classification—precision, recall, F1, ROC-AUC and accuracy when classes are balanced.

3. Core ML Algorithms

Linear Regression

Fits a linear relationship between input features and a continuous target. Useful as an interpretable regression baseline.

Logistic Regression

Uses a logistic function to estimate class probabilities; commonly used for binary classification.

Decision Tree

Learns if/else feature splits. Handles nonlinear relationships and is relatively interpretable, but can overfit.

Random Forest

Combines many decision trees trained on sampled data/features; averaging reduces variance and often improves robustness.

Gradient Boosting

Builds learners sequentially, with later learners correcting earlier errors. Often strong on structured/tabular data.

Support Vector Machine (SVM)

Finds a decision boundary with a large margin; kernels can model nonlinear boundaries.

K-Means

Partitions observations into K clusters by assigning each point to its nearest centroid.

Ensembles

Bagging trains models on samples and combines them; boosting trains sequentially; stacking combines model predictions with a meta-model.

Feature preparation

Handle missing values, encode categories, scale numeric variables when needed, and engineer useful features using domain knowledge.

Key takeaway: Use cross-validation or a validation set for model selection. Choose metrics that match the task and class balance.

4. Neural Networks & Deep Learning

Artificial neuron

Computes z = Σ(w■x■) + b and applies an activation a = f(z). Weights and bias are learned during training.

Network layers

Input layer receives features; hidden layers learn representations; output layer produces predictions or class scores.

Activation functions

ReLU = max(0,x), common in hidden layers. Sigmoid maps to (0,1). Tanh maps to (-1,1). Softmax converts class scores into a probability distribution.

Loss functions

Loss measures prediction error. MSE is common for regression; cross-entropy is common for classification.

Gradient descent

Updates parameters in a direction that reduces loss. Learning rate controls update size.

Backpropagation

Computes gradients through the network using the chain rule so parameters can be updated.

Training terms

Epoch = one pass through training data; batch = subset used for one update; optimizer = rule for updating parameters.

Challenges

Vanishing/exploding gradients, overfitting, high compute requirements and the need for suitable data.

Regularization

Dropout, weight decay, data augmentation and early stopping can reduce overfitting.

Key takeaway: Deep learning is powerful for images, audio and text, but is not automatically better than traditional ML for every tabular dataset.

5. CNNs, Transfer Learning & Quick Review

CNN overview

Convolutional Neural Networks are designed for grid-like data such as images. Filters detect local patterns and produce feature maps.

Convolution terms

Kernel/filter = learned small matrix; stride = movement step; padding = border values that control output size; pooling = downsampling.

Pooling

Max pooling keeps the maximum value in a region; average pooling computes the mean.

LeNet-5

Early CNN architecture used for handwritten digit recognition.

AlexNet

Popularized deeper CNNs, ReLU activations and GPU training at scale.

VGG

Uses repeated stacks of small 3×3 convolution filters.

ResNet

Uses skip/residual connections to help train very deep networks.

Transfer learning

Reuse a pretrained model. Freeze its feature extractor and train a task-specific head; fine-tune layers when more adaptation is needed.

ML vs DL

Traditional ML often needs manual feature engineering and can work well on smaller tabular datasets. DL learns representations and often benefits from more data and compute.

Key takeaway: Quick check: explain AI–ML–DL, supervised/unsupervised/RL, leakage, overfitting, metrics, neuron, backpropagation, convolution and residual connections.